



DOCKER + MONGODB

Running MongoDB in Docker

A local database container, driven entirely from Command Prompt

STEP 1

Start the MongoDB Container

```
//in powershell
docker run -d --name mongodb -p 27017:27017 \
  -e MONGO_INITDB_ROOT_USERNAME=admin \
  -e MONGO_INITDB_ROOT_PASSWORD=admin123 \
  -v mongodb_data:/data/db \
  mongo:8.0
```

-d	Run detached, in the background
--name mongodb	Name the container so later commands can reference it
-p 27017:27017	Map the container's port to the same port on your machine
-e MONGO_INITDB_ROOT_*	Set the root username and password on first boot
-v mongodb_data:/data/db	Persist data in a named volume, safe across restarts
mongo:8.0	The official MongoDB image and version to run

STEP 2

Check Whether It's Running

```
docker ps
```

You should see something like:

```
mongodb      mongo:8.0    ...    0.0.0.0:27017->27017/tcp
```

Nothing listed?

Run `docker ps -a` to see stopped containers too — the run failed silently only if the name `mongodb` is already taken by an older container.

STEP 3

Connect & Explore

Connection string

```
mongodb://admin:admin123@localhost:27017
```

Inside the shell

```
show dbs  
use library_db
```

use creates it lazily

- use library_db switches the active database — it doesn't need to exist yet.
- MongoDB only actually creates it once you write the first document into a collection.
- show dbs won't list library_db until that first insert happens.

Stop & Start the Container

Stop it

```
docker stop mongodb
```

The container shuts down — its data stays intact in the volume.

Start it again

```
docker start mongodb
```

Same container, same data, same credentials — nothing to reconfigure.

stop / start affect the running process only — the container and its named volume both remain on disk in between. Nothing is deleted until you explicitly remove it.

Remove the Container & Its Data

Remove just the container

```
docker rm -f mongodb
```

Deletes the container. The mongodb_data volume — and everything in it — survives.

Start completely fresh

```
docker rm -f mongodb  
docker volume rm mongodb_data
```

Then re-run the original docker run command to get a brand-new, empty database.

Removing the volume is permanent — every collection and document goes with it. There's no undo.

Docker Command Cheat Sheet

<code>docker run ... mongo:8.0</code>	Create and start a new MongoDB container
<code>docker ps</code>	List running containers
<code>docker ps -a</code>	List every container, running or stopped
<code>docker stop mongodb</code>	Stop the container, keep its data
<code>docker start mongodb</code>	Resume a stopped container
<code>docker rm -f mongodb</code>	Delete the container, keep the volume
<code>docker volume rm mongodb_data</code>	Permanently delete the data



KEY TAKEAWAY

The container is disposable — your data lives in the named volume, not the container itself.

Next up: point the CRUD app's MONGO_URI at this container and confirm library_db fills up as you use it.